MIL-D-3945B
3 September 1982
SUPERSEDING
MIL-D-3945A
1 March 1960

MILITARY SPECIFICATION

DETECTOR KIT; CARBON MONOXIDE, COLORIMETRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of portable carbon monoxide detector kit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-P-116 - Preservation, Methods of
MIL-I-3948 - Indicator Tube, Carbon Monoxide
MIL-R-6855 - Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes
MIL-A-8625 - Anodic Coatings, For Aluminum and Aluminum Alloys

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-810 - Environmental Test Methods
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D2227 - Natural Rubber (NR) Technical Grades

(Application for copies should be addressed to ASTM 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Component parts. Each colorimetric carbon monoxide detector kit shall consist of a sampling bulb with metering valve, an indicator tube seal breaker, and a box containing 12 indicator tubes conforming to MIL-I-3948, all contained

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within a carrying case. The box of indicator tubes shall conform to the level A unit package specified in MIL-I-3948. When specified, an extension hose shall be provided (see 6.2).

3.2 Materials.

3.2.1 Nonmagnetic materials. Nonmagnetic materials shall be used.

3.2.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other.

3.2.3 Protective treatment. When materials are used in the construction of the detector kit that are subject to corrosion in salty air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will allow compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided. When aluminum or aluminum alloy parts are used, they shall be anodized in accordance with MIL-A-8625.

3.3 Design and construction.

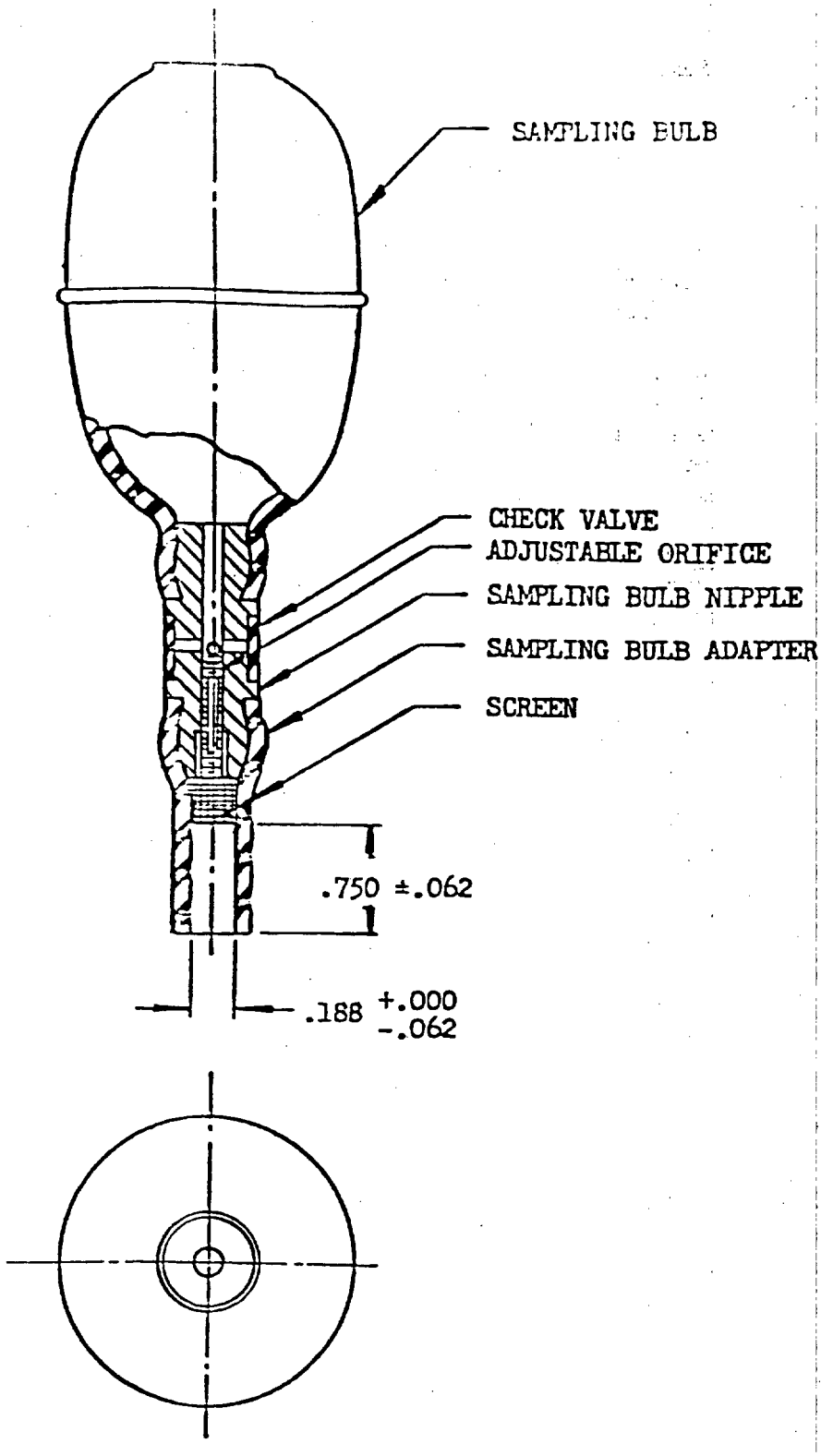
3.3.1 Sampling bulb. The design of the sampling bulb shall conform to figure 1. The volume of the sampling bulb shall be 3.66 $\pm$ 0.30 cubic inches [60 $\pm$ 5 milliliters (ml)]. Sampling bulbs shall be of natural rubber; blue, red, or black; grade optional in accordance with ASTM D2227. The wall thickness shall be no less than 3/32 inch.

3.3.1.1 Sampling bulb nipple. The sampling bulb shall be provided with a metal nipple and shall contain an adjustable orifice and check valve. The design of the sampling bulb nipple shall be similar to that shown in figure 1.

3.3.1.2 Sampling bulb adapter. A tubular adapter of oil-resistant rubber in accordance with MIL-R-6855, class 2, grade 40, and containing a woven filter screen conforming to the internal tubular passage of the adapter, shall be installed on the exposed end of the sampling bulb nipple. Dimensions of the adapter shall conform to figure 1.

3.3.2 Seal breaker. A device for breaking the glass seal at both ends of the indicating tube without rupturing the main body of the indicating tube shall be provided. This device shall include a receptacle for retaining the broken glass seals. The closed end of the seal breaker shall be so designed that it can be used for adjusting the metering valve.

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DIMENSIONS IN INCHES.

FIGURE 1. *Sampling bulb unit*

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3.3.3 Carrying case. A carrying case for storing the sampling bulb, indicator tube seal breaker, calibration and insturction chart, and a carton of 12 tubes shall be provided. The case shall conform to the approximate dimensions shown in figure 2, and the case need not be rectangular in shape. It shall be finished with a durable dull black color. The case shall be of such design as to contain neatly and compactly the sampling bulb, indicator tube seal breaker, calibration and instruction chart, a carton of 12 indicator tubes, and the extension hose with bulb when specified (see 6.2). The carrying case shall be provided with a shoulder strap when specified (see 6.2). When an extension hose is specified (see 6.2), the dimensions of the case may be increased not to exceed 5 inches by 5 inches by 10 inches. A 2-ounce aspirator bulb, similar in general to that specified in 3.3.1, but equipped with a nipple and check valve at each end, shall be attached to one end of the extension hose to serve as a pump for purging the hose before a sample is drawn into the indicator tube.

3.3.4 Extension hose. When specified (see 6.2), an oil-resistant rubber hose shall be provided with each detector kit for the purpose of remote sampling. The hose shall be 3/16-inch inner diameter, approximately 5/16-inch outer diameter, and 10 feet long, and shall be in accordance with MIL-R-6855, class 2, grade 40.

3.3.5 Carrying strap. A carrying strap of good quality leather or webbing shall be provided with the carrying case when specified (see 6.2).

3.4 Weight. The weight of the complete detector kit, less the extension hose and carrying strap, shall not exceed 1.5 pounds.

3.5 Check valve leakage. The outlet valve shall not leak when tested as specified in 4.4.4.1.

3.6 Low temperature operation. The detector kit shall operate satisfactorily when tested as specified in 4.4.4.2.

3.7 High temperature operation. The detector kit shall operate satisfactorily when tested as specified in 4.4.4.3.

3.8 Inflation time. The inflation time of the bulb, without the indicator tubes shall be 28 $\pm$ 3 seconds when tested as specified in 4.4.4.4.

3.9 Marking. Equipment, assemblies, and parts, shall be marked for identification in accordance with MIL-STD-130.

3.10 Workmanship. The detector kit, including all parts and accessories, shall be uniform in quality and shall be free from irregularities which would adversely affect the appearance or serviceability of the detector kit. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

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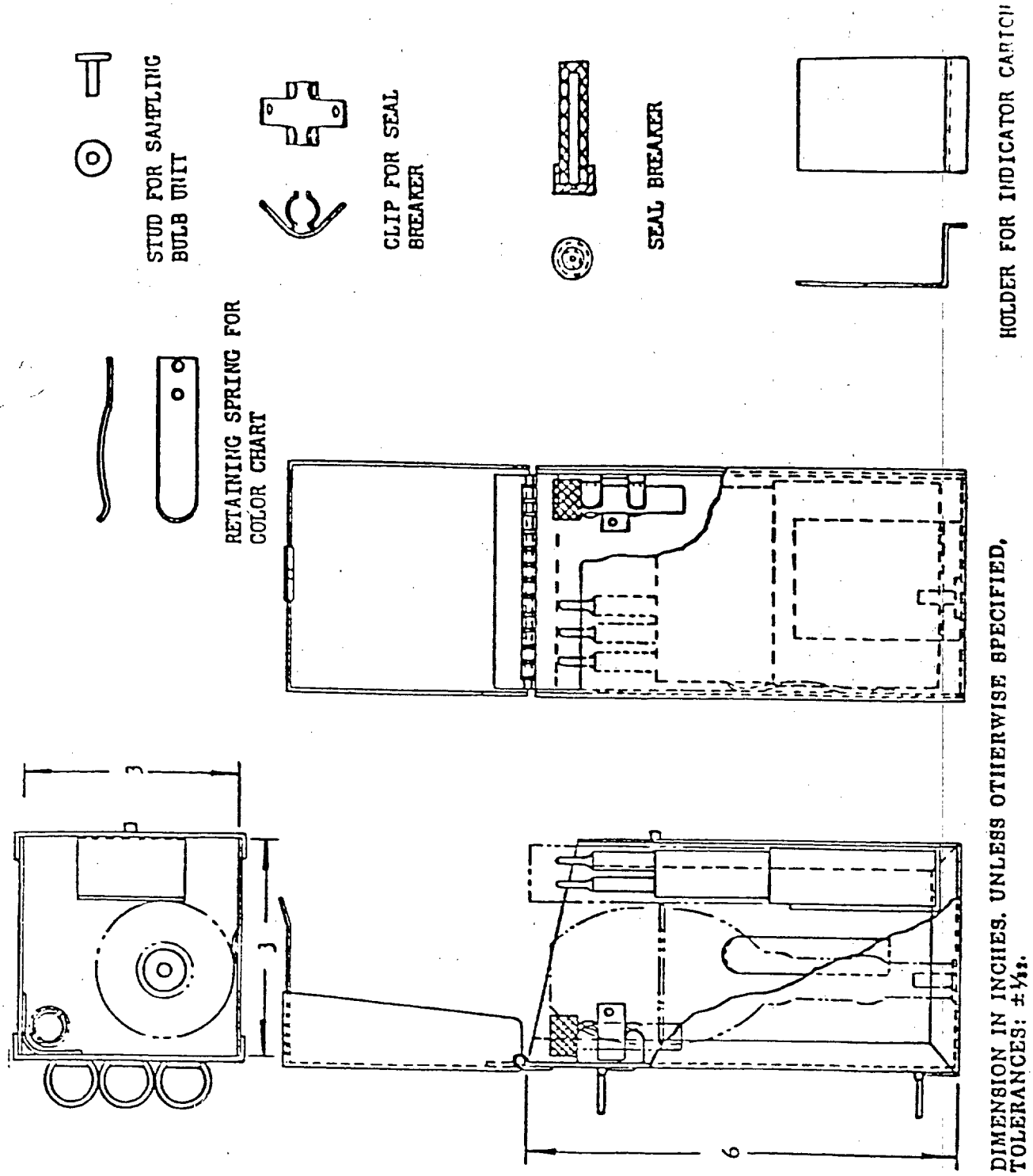


FIGURE 2. *Carrying case for carbon monoxide indicator bulb-type colorimetric*

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3.11 Preproduction. Prior to the start of regular production, a preproduction sample quantity of detector kits shall be produced in accordance with this specification for examination and test (see 4.3).

3.11.1 Low temperature. There shall be no damage which would adversely affect subsequent operation when the detector kit is tested as specified in 4.3.3.1(a).

3.11.2 High temperature. The detector kit shall operate satisfactorily when tested as specified in 4.3.3.1(b).

3.11.3 Humidity. There shall be no evidence of corrosion or rusting of the internal or external parts of the detector kit that might affect subsequent operation when the detector kit is tested as specified in 4.3.3.1(c).

3.11.4 Fungus resistance. The detector kit shall not support fungus growth when tested as specified in 4.3.3.1(d).

3.11.5 Salt spray. There shall be no evidence of corrosion or rusting of the internal or external parts of the detector kit that might affect subsequent operation when the detector kit is tested as specified in 4.3.3.1(e).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of no less than 100 detector kits shall be produced using the same methods, materials, equipment, and processes of manufacture as will be used during regular production. The preproduction lot shall contain no more than one lot of each component.

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4.3.2 Sampling and inspection procedure.

4.3.2.1 For examination of end item. Sample preproduction detector kits shall be examined in accordance with 4.4.2.1 and 4.4.3.1.

4.3.2.2 For examination of packaging. Sample preproduction detector kits shall be examined in accordance with 4.4.2.2 and 4.4.3.2.

4.3.2.3 For test. Each detector kit in the preproduction lot shall be tested as specified in 4.4.4.1. Failure of a detector kit to pass the test specified in 4.4.4.1 shall be cause for rejection of the tested detector kit. Five sample detector kits shall be randomly selected from the preproduction lot and tested as specified in 4.3.3 and 4.4.4, excluding 4.4.4.1. Failure of any preproduction test or quality conformance test excluding 4.4.4.1, by any detector kit, shall be cause for rejection of the preproduction lot.

4.3.3 Preproduction tests. Tests shall be conducted as follows:

4.3.3.1 Environmental. Subject the detector kit to the following tests in accordance with MIL-STD-810, except as specified below. At the conclusion of each test and after it has returned to room temperature, the detector kit shall meet the requirements specified in 4.4.4.4. No damage shall result from each test.

(a) Low temperature. Subject the detector kit to low temperature test method 502, procedure I, of MIL-STD-810 except that -4°F (-20°C) shall be substituted for -70°F (-57°C).

(b) High temperature. Subject the detector kit to high temperature test method 501, procedure I, of MIL-STD-810, except that the period of exposure shall be 24 hours.

(c) Humidity. Subject the detector kit to humidity tests in accordance with method 507, procedure V of MIL-STD-810, except that the temperature and relative humidity in the chamber shall be +49°C (+120°F) and 95 percent, respectively.

(d) Fungus resistance. Subject the detector kit to fungus resistance test method 508, procedure I, of MIL-STD-810.

(e) Salt spray. Conduct the salt spray in accordance with test method 509, procedure I, of MIL-STD-810.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the detector kits produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions.

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4.4.2 Sampling.

4.4.2.1 For examination of end item. Five detector kits shall be randomly selected from each lot.

4.4.2.2 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.3 For tests. Each detector kit shall be tested as specified in 4.4.4.1. In addition, five detector kits shall be randomly selected from each lot.

4.4.3 Inspection procedure.

4.4.3.1 For examination of end item. Sample detector kits and components selected in accordance with 4.4.2.1 shall be examined in accordance with the following classification of defects. Any failure by any detector kit will be cause for rejection of the lot represented.

(a) Components or component parts damaged, missing, or incorrect (as required)
(b) Protective coating cracked, chipped, or scaled (when required)
(c) Components or component parts incorrectly assembled
(d) Box of indicator tubes not packaged as specified (see 3.1)
(e) Weight of kit not as specified (see 3.4)
(f) Marking of parts not as specified (see 3.8)
(g) Workmanship not as specified (see 3.9)

4.4.3.2 For examination of packaging. The sample unit shall be one unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping container shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged
(e) Cleaning and drying not as specified
(f) Filler pads missing or not as specified (when required)
(g) Box size and weight not as specified
(h) Strapping missing or not as specified (when required)
(i) Container marking missing, illegible, or not as specified

4.4.3.3 For test. All kits shall be tested as specified in 4.4.4.1. Failure of a kit to pass the test specified in 4.4.4.1 shall be cause for rejection of the tested detector kit. The five kits selected in accordance with 4.4.2.3 shall be tested as specified in 4.4.4.2, 4.4.4.3, and 4.4.4.4. Failure of any test by any kit shall be cause for rejection of the lot represented.

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4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Check valve leakage. Insert a solid plug or unopened indicator tube into the bulb adapter and deflate the bulb to completely expel the air. There shall be no detectable inflation of the bulb in a 1 minute period.

4.4.4.2 Low temperature operation. Subject the detector kit to a temperature of -4°F (-20°C) for a period of 4 hours. At the end of this period, and while still at the low temperature, the detector kit shall pass the tests specified in 4.4.4.1 and 4.4.4.4. After the detector kit has returned to room temperature, it shall pass the tests specified in 4.4.4.1 and 4.4.4.4.

4.4.4.3 High temperature operation. Place the detector kit in a chamber maintained at a temperature of +160°F (+71°C) for a period of 4 hours. It shall pass the tests specified in 4.4.4.1 and 4.4.4.4 while still at this temperature. After the detector kit has returned to normal temperature, it shall pass the tests specified in 4.4.4.1 and 4.4.4.4.

4.4.4.4 Inflation time. Deflate the bulb, less the indicator tube, to exclude air as completely as possible, and then allow it to refill. The time required to refill the bulb shall be 28 ± 3 seconds. The orifice of the bulb may be adjusted under any atmospheric conditions to permit inflation of the bulb within the specified time.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or industrial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Components of the detector kit shall be cleaned in accordance with method C-1 of MIL-P-116, and dried in accordance with any applicable method listed therein.

5.1.1.2 Unit packing. The detector kits, cleaned and dried in accordance with 5.1.1.1, shall be unit packed, one per package, in accordance with method IA-8 of MIL-P-116. Preservative is not required. Included air volume in the sealed package shall be held to a minimum.

5.1.1.3 Intermediate packing. The detector kits, unit packed in accordance with 5.1.1.2, shall be intermediate packed within a box conforming to type CF, class weather-resistant, variety SW, grade optional of PPP-B-636. Intermediate containers shall contain like quantities and shall not exceed the size and weight limitations of the fiberboard box. Filler pads conforming to class weather-resistant, variety SW, grade optional of PPP-F-320, shall be used as necessary. Not more than two pads per interior face of any one box shall be

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used to protect the unit packages from damage by metal fasteners or restrict free movement of the unit packages. The box shall be closed and sealed in accordance with the closure requirements specified in the appendix of PPP-B-636 for weather-resistant boxes.

5.1.2 Level B.

5.1.2.1 Unit packing. The detector kits shall be unit packed in accordance with 5.1.1.2.

5.1.2.2 Intermediate packing. The detector kits, unit packed in accordance with 5.1.2.1, shall be intermediate packed in accordance with 5.1.1.3, except the intermediate box shall be type CF, class domestic, variety SW, grade 175 of PPP-B-636. Boxes shall be closed and sealed in accordance with the closure requirements specified in the appendix of PPP-B-636 for domestic boxes.

5.1.3 Industrial packing. Each detector kit shall be preserved and packed in accordance with the requirements specified in MIL-STD-1188.

5.2 Packing. Packing shall be levels A, B or industrial as specified (see 6.2).

5.2.1 Level A. Detector kits, prepared in accordance with 5.1.1, shall be packed within a box conforming to either overseas type, style A, grade A of PPP-B-601; or class 2, style 4, grade A of PPP-B-621. Each box shall contain like quantities and shall not exceed the box size limitations and shall not exceed a gross weight of 65 pounds. Filler pads conforming to weather-resistant, variety SW, grade optional of PPP-F-320 shall be used as necessary to restrict free movement of the packages. The box shall be closed and strapped in accordance with the closure requirements specified in the appendix to PPP-B-601 or PPP-B-621, as applicable. Strapping shall conform to type I, class 1 of QQ-S-781.

5.2.2 Level B. Detector kits, prepared as specified in 5.1.2, shall be packed within a fiberboard box conforming to type CF, class weather-resistant, variety SW, grade V3c of PPP-B-636. The boxes shall contain like quantities and shall not exceed either the size or the weight limitations of the fiberboard box. Filler pads conforming to class weather-resistant, variety SW, grade V3c of PPP-F-320 shall be used as necessary to restrict free movement of the packages within the shipping container. The box shall be closed and sealed in accordance with the closure requirements specified in the appendix of PPP-B-636 for weather-resistant boxes. The box shall be reinforced with tape banding in accordance with the requirements specified in the appendix of PPP-B-636.

5.2.3 Industrial packing. Detector kits industrially packed as specified in 5.1.3, shall be packed in accordance with MIL-STD-1188.

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5.3 Marking. Marking for levels A and B unit packing, intermediate packing, and packing shall be in accordance with MIL-STD-129. Marking shall be in accordance with MIL-STD-129. Marking for industrial preservation and packing shall be in accordance with MIL-STD-1188. In addition, the month and year of expiration of the inclosed detector tube component shall be shown to the right of the identification marking on each shipping container as follows:

INCLOSED INDICATOR TUBES
EXPIRES: (Insert applicable month and year).

6. NOTES

6.1 Intended use. The detector kit covered by this specification is intended for use to detect and evaluate the concentration of carbon monoxide.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of preservation and packing required (see 5.1 and 5.2)
(c) If shoulder strap is required (see 3.4.3)
(d) If extension hose is required (see 3.2)
(e) Preproduction instructions

Custodians:

Army - EA
Air Force - 99
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Review activities:

Army - MD
Navy - MS
DLA - DM

Preparing activity:

Army - EA

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